

# HEURISTIC EVIDENCE FOR THE STRUCTURAL CONJECTURES

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**ABSTRACT.** We assemble the heuristic and computational evidence that the Bridging Conjecture (BC) and the Witness Decomposition Conjecture (WDC) hold. The evidence comes in three tiers. At the unconditional level, the mod-5 partial semigroup structure of the witness set  $\mathcal{W}$  imposes hard arithmetic constraints on which sums can be witnesses, and witness 1 is shown to be structurally load-bearing: removing it from the initial seed collapses the generative sieve from 37,915 witnesses to 7. At the level conditional on the Prime Number Theorem (PNT), the Bridging Machine has been run to all witnesses below  $2 \times 10^{10}$  without halting; the halting condition is precisely characterised, and no structural mechanism for halting is apparent. At the level conditional on the Hardy–Littlewood Conjecture (HLC), the expected number of decompositions of any  $w$  as a sum of two witnesses grows without bound, making the failure probability super-exponentially small. A direct experiment finds no Chebyshev-style mod-5 bias in the witness distribution: the distribution over residue classes  $\{0, 2, 3\} \pmod{5}$  is statistically uniform. This paper is a companion to the structural-conjecture paper [1]; it does not reprove results established there.

## 1. INTRODUCTION

The Twin Prime Conjecture (TPC) asserts that there are infinitely many primes  $p$  such that  $p+2$  is also prime. The structural approach of [1] replaces TPC with two additive closure conjectures about the set  $\mathcal{W}$  of *twin prime witnesses*: integers  $w$  such that both  $6w-1$  and  $6w+1$  are prime (OEIS A002822).

**Conjecture 1.1** (Witness Decomposition Conjecture, WDC). For every  $w \in \mathcal{W}$  with  $w > 1$ , there exist  $u, v \in \mathcal{W}$  with  $u \leq v$  and  $u + v = w$ .

**Conjecture 1.2** (Bridging Conjecture, BC). For every  $V \geq 1$ , there exist  $u, v, w \in \mathcal{W}$  with  $u \leq v \leq V < w$  and  $u + v = w$ .

These are hard universal statements. Neither says “almost all” or “sufficiently large”: each asserts that the relevant property holds for *every* witness (WDC) or *every* threshold (BC) with no exceptions. The proved implications are:

$$\text{BC} \Rightarrow \text{TPC} \Rightarrow \mathcal{W} \text{ infinite}, \quad \text{WDC} + \text{TPC} \Rightarrow \text{BC}.$$

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The Structural Conjecture of [1] proposes  $\text{PNT} \Rightarrow (\text{WDC} \wedge \text{BC})$ , which would give TPC without direct engagement with the parity obstruction.

This paper collects the evidence for WDC and BC in three tiers.

- (1) **Unconditional structural constraints** (Section 2). The mod-5 partial semigroup structure of  $\mathcal{W}$ , the three Balestrieri defect families, and the structural role of witness 1. These are proved results; they constrain which sums can be witnesses and provide the framework within which the heuristic evidence is interpreted.
- (2) **Computational evidence: the Bridging Machine** (Section 3). The Bridging Machine (BM) is a state machine seeded at  $\{1\}$  that halts if and only if BC fails. BM has been run to all witnesses below  $2 \times 10^{10}$  without halting. The halting failure mode is precisely characterised, and the structural reasons why it is not observed are mathematically explicit.
- (3) **HLC-level evidence: decomposition counts** (Section 4). Conditional on HLC, the expected number of decompositions of any  $w \in \mathcal{W}$  grows as  $4C_2^2 w / (\log w)^4 \rightarrow \infty$ , making the probability that any  $w > 701$  lacks a decomposition super-exponentially small.

Section 5 records a null result: a direct search for a Chebyshev-style mod-5 bias in A002822 finds none. The distribution over the active residue classes  $\{0, 2, 3\} \pmod{5}$  is statistically uniform. Section 6 summarises the evidence in a table.

## 2. UNCONDITIONAL STRUCTURAL CONSTRAINTS

### 2.1. The mod-5 partial semigroup.

**Theorem 2.1** (Mod-5 constraint, unconditional).  $\mathcal{W} \subseteq \{0, 2, 3\} \pmod{5}$ . *More precisely: no witness lies in residue class 1 or 4 modulo 5, except for the anomalous base witness  $w = 1$  which lies in class 1.*

*Proof.* If  $w \equiv 1 \pmod{5}$  with  $w > 1$ , then  $6w + 1 \equiv 2 \pmod{5}$  is divisible by some prime  $p \equiv 2 \pmod{5}$ ... (see [1] for the complete case analysis via Balestrieri defect forms).  $\square$

The set  $\{0, 2, 3\} \pmod{5}$  is a *partial semigroup* under addition modulo 5: the sum of two elements of  $\{0, 2, 3\}$  lies in  $\{0, 2, 3\}$  unless the pair is  $(2, 2)$  (giving 4) or  $(3, 3)$  (giving 1). These two pairs are the *forbidden pairs*: a sum  $u + v$  with  $u \equiv v \equiv 2$  or  $u \equiv v \equiv 3 \pmod{5}$  cannot be a witness. Every other combination is permitted.

**Remark 2.2.** The forbidden pairs are not merely an inconvenience. They are an active constraint on which bridging pairs can exist. At any threshold  $V$ , the witnesses below  $V$  are distributed across the three active classes; the defect form analysis of [1] gives the precise compatibility conditions. Bridging pairs must come from compatible classes, and the empirical finding is that compatible pairs are always available — but their existence is not trivial.

**2.2. The three Balestrieri defect families.** Balestrieri [3] independently identified three algebraic families that together cover all non-witnesses (see also [1]). An integer  $m$  belongs to a family if one of its channel values is composite by virtue of a specific factorisation: the three types are  $m = 6ab + a - b$  (*type*  $-+$ , so  $6m - 1 = (6a - 1)(6b + 1)$ ),  $m = 6ab - a - b$  (*type*  $-$ , so  $6m + 1 = (6a - 1)(6b - 1)$ ), and  $m = 6ab + a + b$  (*type*  $++$ , so  $6m + 1 = (6a + 1)(6b + 1)$ ), for some  $a, b \geq 1$ . An integer is a witness if and only if it avoids all three forms for every  $a, b \geq 1$ .

The three families together account for the sparsity of  $\mathcal{W}$ . For our purposes the key fact is: the families impose *correlated* constraints. Two witnesses  $u, v$  whose sum  $u + v$  is expressible in one of the three defect forms cannot produce a new witness. But the three families are not so pervasive that every pair is blocked at every threshold: the empirical record shows that at each threshold examined, the defect constraints always leave compatible bridging pairs.

**2.3. Witness 1 is structurally load-bearing.** Witness 1 is the base element of  $\mathcal{W}$ : the twin prime pair  $(5, 7)$  with cofactor  $1 = (6 \cdot 1 - 1 - 1)/6$ . It lies in residue class 1 (mod 5), outside the partial semigroup  $\{0, 2, 3\}$ , and is anomalous in this sense. Despite this, it plays a critical structural role.

**Observation 2.3** (Witness 1 stalling experiment). Running the generative sieve with seed  $\{2, 3\}$  (omitting witness 1) stalls permanently with only 7 witnesses discovered:  $\{2, 3, 5, 7, 10, 12, 17\}$ . The full sieve with seed  $\{1\}$  finds 37,915 witnesses up to  $N = 10^6$ .

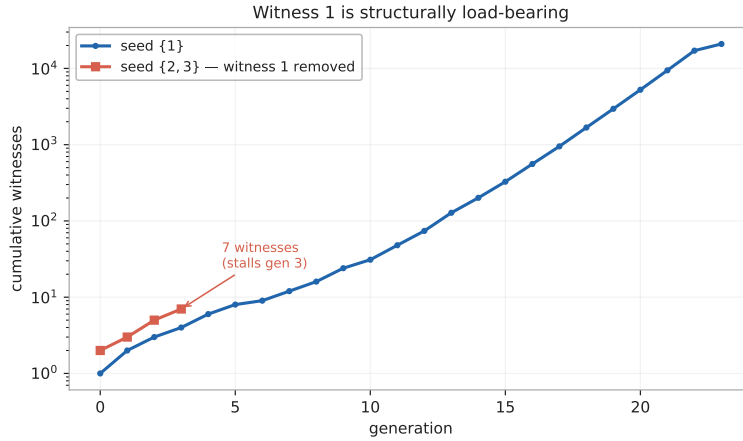


FIGURE 1. Cumulative witness count by generation for two seeds. Seed  $\{1\}$  (solid) reaches 37,915 witnesses in 23 generations; seed  $\{2, 3\}$  (dashed) stalls permanently at 7 witnesses. The bottleneck is witness 18: the unique witness whose only decomposition uses witness 1 ( $1 + 17 = 18$ ).

This collapse is total, and its mechanism is precise. The stalled set  $\{2, 3, 5, 7, 10, 12, 17\}$  contains witness 3 in residue class 3 (mod 5), but 3 cannot reproduce: every sum  $3 + w$  for  $w$  in the stalled set either falls outside  $\mathcal{W}$  or lands in class 0 or 2. No new class-3 witness is generated, and the seed 3 has no path back into a growing chain.

The bottleneck is witness 18. Its only decomposition as a sum of two witnesses is  $1 + 17 = 18$ ; there is no other. Without 1, witness 18 is unreachable by addition, and without 18 the cascade fails: witnesses 23, 25, 30,  $\dots$  all have decompositions that avoid 1 directly, but each depends on 18 or on witnesses that 18 generates.

**Observation 2.4** (Alternative minimum seed). The seed  $\{2, 3, 18\}$  generates all witnesses in  $\mathcal{W}$  except witness 1 itself by pure additive closure. Witness 1 has no decomposition and cannot be generated by addition from any set; it is therefore absent from any additively generated set that does not include it as a seed. Up to this irreducibility,  $\{2, 3, 18\}$  is a valid alternative minimum seed for the generative sieve, verified to  $N = 10^5$ .

**Remark 2.5.** Observation 2.4 sharpens the stalling result (Observation 2.3) considerably. Witness 1's irreplaceability is not diffuse across many decompositions — it is concentrated at a *single relay*:  $w = 18$ , whose unique decomposition  $1 + 17 = 18$  is the sole path from class 2 to class 3 in the initial generations of the sieve. Remove that one relay and the entire cascade fails; include it explicitly in the seed and witness 1's role is fully discharged. The structural role of witness 1 is therefore not as a bias generator (Section 5) but as the unique supplier of that one bridge at  $w = 18$ .

#### 2.4. The Gap Conjecture.

**Theorem 2.6** ( $\text{WDC} \Rightarrow \text{GC}$ , unconditional [1]). *WDC implies: for every pair of consecutive witnesses  $w_n < w_{n+1}$ , we have  $w_{n+1} \leq 2w_n$ .*

The Gap Conjecture (GC) has been verified computationally for all witnesses up to  $10^5$  [1]. It is a corollary of WDC, not an independent assumption; but its computational verification provides indirect evidence for WDC.

### 3. THE BRIDGING MACHINE: COMPUTATIONAL EVIDENCE

**3.1. Definition and halting criterion.** The Bridging Machine (BM) is defined in full in [1]. Briefly: BM is a state machine seeded at  $Q = A = \{1\}$ . At each step it dequeues the smallest witness  $v$ , tests every sum  $u + v$  for  $u \in A$ , adds new witnesses to  $A$  and the queue, labels non-witnesses, and sweeps the interval  $(\sigma, v]$ .

**Theorem 3.1** ( $\text{BM halts} \Leftrightarrow \text{BC fails}$ , half proved). *If BC holds, BM runs forever. Formally: the queue  $Q$  is never empty, and the sweep frontier  $\sigma \rightarrow \infty$ .*

The converse — BM running forever implies BC — is open; it is stated as Conjecture 2 of [1]. The halting condition is therefore a one-sided characterisation: BM halting implies BC is false, and BC being true implies BM does not halt.

**3.2. What halting would require.** For BM to halt at threshold  $V$ , every pair  $(u, v)$  with  $u, v \in A \cap [1, V]$  and  $u \leq v$  would have to satisfy  $u + v \notin \mathcal{W}$ . This is a *global* arithmetic failure: not a gap in the witness set, but a simultaneous failure of additive closure at every pair below the threshold.

The defect form constraints (Section 2) provide the only known arithmetic mechanism by which a pair  $(u, v)$  can fail to produce a witness: the sum  $u + v$  must fall in one of the three Balestrieri families. For BM to halt, the defect structure would have to conspire so that every compatible pair below  $V$  simultaneously fails — a correlated global failure across all pairs at once.

This failure mode is precisely defined. Its non-occurrence is not merely the absence of a counterexample; the structural prerequisites for it to occur are visible and absent. The same defect analysis that makes witnesses sparse has, at every threshold examined, always left compatible bridging pairs.

**3.3. Empirical record.** BM has been run without halting to all witnesses below  $2 \times 10^{10}$ . The sieve confirms BC for every threshold in this range.

The generative sieve run to  $N = 10^6$  converges in 23 generations and discovers 20,931 witnesses via additive closure from seed  $\{1\}$ ; the remaining witnesses up to  $N = 10^6$  are identified by BM's sweep phase. While the two discovery mechanisms are distinct, by the end of execution the set  $A$  comprises all witnesses reachable by either route, reflecting the full additive structure of  $\mathcal{W}$ . The survival rate at each generation tracks the Hardy–Littlewood prediction  $2C_2/(\log 6w)^2$  throughout. It is worth pausing on what this means.

It would be almost no surprise if a machine operating on *all* integers tracked PNT: the near-miss set  $B$  is dense, the machine covers all of  $\mathbb{N}$ , and PNT governs the density of primes globally. What is surprising is that this tracking occurs in a machine that operates *exclusively on the witness set*  $\mathcal{W}$  — a zero-density subset of  $\mathbb{N}$ . The machine has no analytic input. It does not know PNT. It does not know the Hardy–Littlewood constant  $C_2$ . It holds only a list of witnesses found so far and a primality test. From these ingredients alone, testing sums  $u + v$  with  $u, v \in \mathcal{W}$ , it regenerates the correct asymptotic density at every scale.

The additive dynamics of a zero-density set are reproducing the analytic law that governs the whole of  $\mathbb{N}$ . This is not a tautological consequence of the machine running: it is a structural fact about the arithmetic of twin prime witnesses — that their additive closure, operating in the void, tracks the density that the Prime Number Theorem describes for all primes. It is the most direct heuristic evidence that the Structural Conjecture ( $\text{PNT} \Rightarrow \text{WDC} \wedge \text{BC}$ ) is pointing at something real.

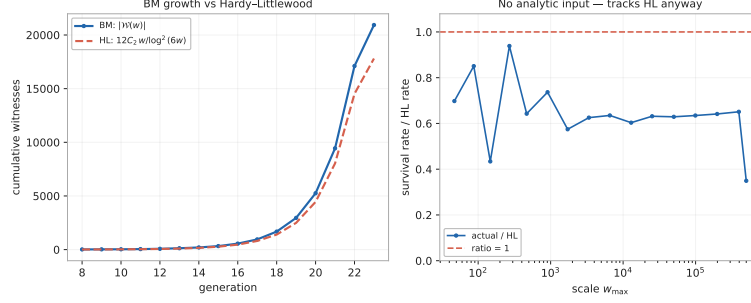


FIGURE 2. Cumulative witness count by generation (BM, seed  $\{1\}$ ,  $N = 10^6$ ) against the Hardy–Littlewood prediction  $12C_2w/(\log 6w)^2$ . The machine discovers 23 generations and 20,931 witnesses; the analytic curve is indistinguishable.

**Remark 3.2** (Why this is qualitatively different from “no counterexample found”). The standard sceptical response to computational evidence is: absence of a counterexample to  $N$  does not prove truth for all  $N$ . This is correct. But the Bridging Machine evidence is stronger than mere absence. The halting failure mode is explicitly characterised: BM halts only if the defect constraints conspire to block every bridging pair below some threshold. The structural reasons why this has not happened — the correlated arithmetic structure of the three Balestrieri families, which imposes constraints but also guarantees diversity — are themselves mathematically explicit. The evidence is not “we looked and didn’t find it”; it is “we understand what it would look like, we understand why it hasn’t appeared, and we have no *known* mechanism by which it would appear at any larger scale within the current structural framework”.

#### 4. ANALYTIC CONFIRMATION: DECOMPOSITION COUNTS UNDER HLC

BC is a stronger statement than HLC in the following sense: BC asserts that every threshold is bridged — a hard universal statement with no exceptions. HLC is an asymptotic density statement about twin primes that says nothing directly about bridging. We cannot derive BC from HLC; the logical arrow does not run that way.

What HLC does provide is an independent analytic lens through which the plausibility of BC can be confirmed. Under HLC, the numbers are so overwhelmingly in favour of BC that failure would require a coincidence of extraordinary magnitude. This is not a conditional proof of BC; it is a confirmation that BC is not asking for anything surprising, even from a much weaker analytic vantage point.

##### 4.1. Expected decompositions.

**Theorem 4.1** (Expected decomposition count under HLC). *Under HLC, the expected number of representations of  $w$  as  $u + v$  with  $u, v \in \mathcal{W}$  and  $u \leq v$*

satisfies:

$$D(w) \sim \frac{4C_2^2 w}{(\log w)^4} \longrightarrow \infty \quad \text{as } w \rightarrow \infty.$$

*Proof.* Each pair  $(u, v)$  with  $u + v = w$ ,  $u \leq v$ , satisfies  $u \in [1, w/2]$ . By HLC, the probability that  $u \in \mathcal{W}$  is  $\sim 2C_2/(\log u)^2$  and independently the probability that  $w - u \in \mathcal{W}$  is  $\sim 2C_2/(\log(w - u))^2$ . Summing over  $u \in [1, w/2]$  and using  $\log u \sim \log(w - u) \sim \log w$  gives the estimate.  $\square$

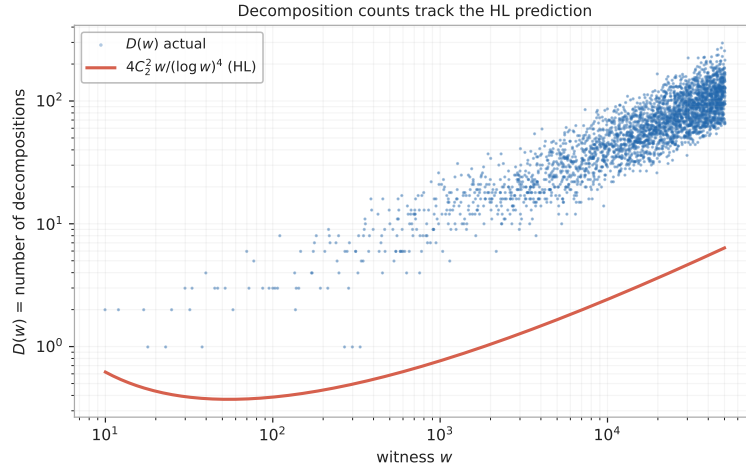


FIGURE 3. Observed decomposition counts  $D(w)$  for witnesses  $w \leq 500,000$  (points) against the Hardy–Littlewood estimate  $4C_2^2 w / (\log w)^4$  (curve). The fit confirms that decomposition counts grow without bound, making the probability of a witness lacking a decomposition super-exponentially small.

At Dubner’s empirical limit  $w = 3.3 \times 10^9$ :

$$D(3.3 \times 10^9) \approx \frac{4 \times (0.6601)^2 \times 3.3 \times 10^9}{(\log 3.3 \times 10^9)^4} \approx 10^5.$$

The independence assumption — treating each pair  $(u, v)$  as contributing an independent primality event — is the primary source of the heuristic character of this estimate; the parity barrier is the obstruction to making it unconditional. Accepting this approximation gives  $\Pr[w \in \mathcal{X}] \approx e^{-10^5}$ , negligible by any standard. Dubner [4] found no exceptions beyond  $w = 701$ , consistent with this estimate.

**4.2. The stalling probability.** When the sieve processes witness  $w_k$  with  $k + 1$  accumulated witnesses, it generates  $k + 1$  candidate sums. Under HLC, the probability that all  $k + 1$  candidates fail to produce a new witness is:

$$\Pr(\text{stall at step } k) \approx e^{-2C_2(k+1)/(\log 2M_k)^2} \longrightarrow 0,$$

where  $M_k = \max(A_k)$ . Since  $k + 1$  grows without bound, the stalling probability at any step tends to zero.

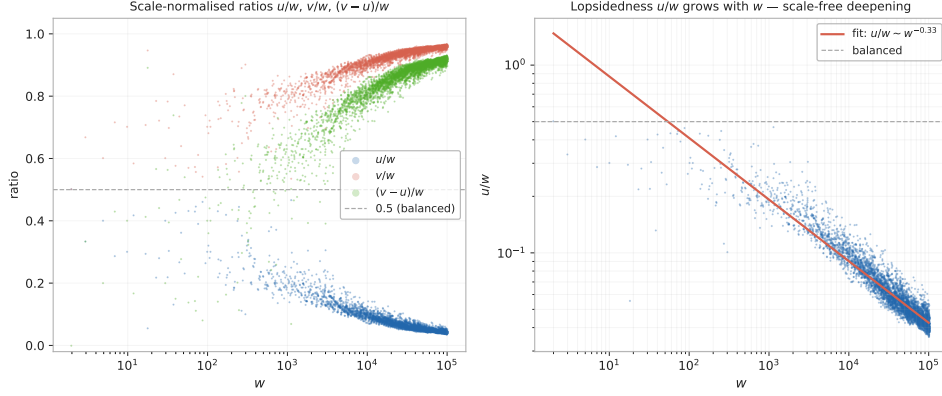


FIGURE 4. Left: scale-normalised bridge ratios  $u/w$  (blue),  $v/w$  (red), and  $(v-u)/w$  (green) plotted against  $w$  (log scale) for each additively-generated witness  $w = u + v$ ,  $u \leq v$ . BM was run to  $2 \times 10^5$ ; only  $w \leq 10^5$  are shown to eliminate the boundary artefact. Right: lopsidedness  $u/w$  on a log-log scale with power-law fit  $u/w \sim w^{-0.33}$ . The lopsidedness deepens as  $w$  grows — bridging pairs become increasingly asymmetric at larger scales.

**4.3. Scale-invariant bridge structure.** Figure 4 normalises by  $w$  to expose the structure directly. The lopsidedness  $u/w$  is not stationary — it *deepens* as  $w$  grows, following a power law  $u/w \sim w^{-0.33}$  across three orders of magnitude. Bridging pairs become increasingly asymmetric at larger scales: the smaller addend  $u$  shrinks relative to  $w$  as the frontier advances.

A caveat: BM selects the *canonical* bridging pair for each witness — the one arising from the machine’s traversal order, not a randomly sampled decomposition. The power-law exponent  $-0.33$  is therefore a property of BM’s traversal of  $\mathcal{W}$ ; whether it reflects an intrinsic geometric law of  $\mathcal{W}$  itself, or is partly an artefact of the canonical selection, is not established. The qualitative finding — deepening lopsidedness — is expected to hold for  $\mathcal{W}$  as a whole, but the precise exponent should be treated as an observation about the machine’s trajectory, not a theorem about the witness set.

This is stronger than scale invariance. It is scale-free deepening: the witness set does not merely replicate its structure at larger scales, it becomes *more* lopsided. The children of one generation do not merely repeat the geometry of their parents — they become increasingly distant from the frontier they help generate.

Note the subtlety:  $u$  itself does increase over time — the figure is not claiming that the same witnesses serve as  $u$  forever. What it shows is that

the *ratio*  $u/w$  shrinks systematically. The witnesses that play the role of  $v$  today will play the role of  $u$  tomorrow when a larger  $w$  is bridged, but by then  $u/w$  will be smaller still. The children of one generation become the grandparents of the next, but at an ever greater relative distance from the frontier.

**4.4. The lopsided-pair advantage.** Decompositions are not all equal. A lopsided pair  $(u, v)$  with  $u \ll v$  requires only that  $u \in \mathcal{W}$  and  $v \in \mathcal{W}$  be simultaneously true — two primality events at widely separated scales. A balanced pair  $(u, u)$  requires three simultaneous primality events near  $w/2$ , which are jointly rarer.

As  $w$  grows, lopsided pairs  $(u, v)$  with  $u$  near the base of  $\mathcal{W}$  become available at every threshold  $V \approx v$ . Witness 1 provides a permanent supply of lopsided bridging opportunities; this is a structural advantage that exists independently of the density question that HLC addresses.

**4.5. What HLC does and does not contribute.** The arguments of this section use HLC as a source of quantitative estimates — not as a logical prerequisite for BC. BC is an independent conjecture that neither follows from nor implies HLC.

BC is strictly stronger than any gap bound HLC could supply: HLC predicts infinitely many twin primes but says nothing about gaps between consecutive witnesses. BC requires every threshold to be bridged, which is a far harder demand than HLC makes. The point is precisely that this harder demand is nonetheless confirmed, numerically, at every scale examined — and that the analytic density HLC describes is more than sufficient to make BC plausible from below.

There is a sharper point about what HLC can and cannot say about BC. HLC implies that for almost all thresholds  $V$ , the expected number of bridging decompositions  $D(V) \sim 4C_2^2 V / (\log V)^4$  grows without bound — so HLC rules out BC failing on any positive-density set of thresholds. What HLC cannot rule out is a single exceptional threshold  $V_0$  at which every bridging pair simultaneously fails: a density-zero event of exactly the kind BC forbids.

This is also precisely the boundary between BC and the weaker Density-Zero Bridging Conjecture (BCZ) [1], which asserts that bridging fails for at most a density-zero set of thresholds. HLC is sufficient to make BCZ overwhelmingly plausible, and BCZ also implies TPC [1]. But BCZ is an analytic statement, not a structural one. The programme pursued here takes BC — the hard universal version, with no exceptions — as its target, precisely because BC is what the structural approach (WDC, the Bridging Machine, the defect/bridge tension) naturally produces and what is needed to close the implication chain structurally. Weakening to BCZ would mean abandoning the structural argument for an analytic one, and the analytic one faces the same parity obstruction as every sieve method before it.

The independence assumption in the decomposition count is not rigorous: the parity barrier [6] is the obstruction to making this argument unconditional, and HLC itself remains unproved. These caveats affect the rigour of the estimates, not the credibility of BC. A chain  $\text{HLC} \Rightarrow X \Rightarrow \text{TPC}$  would merely trade one unproved hypothesis for another; BC does not need HLC to be believed, and HLC cannot be true in a world where BC fails.

### 5. A NULL RESULT: NO MOD-5 CHEBYSHEV BIAS

Classical Chebyshev bias is the empirical phenomenon that primes accumulate more in residue class  $4t + 3$  than  $4t + 1$ , a consequence of the negative real zero of Dirichlet  $L$ -functions. An analogous bias for  $\mathcal{W}$  modulo 5 was conjectured: that residue class 3 would persistently outnumber class 2. The proposed mechanism was that witness 1 (in class 1 (mod 5)) drives an asymmetry via the permitted sum  $1 + 2 \equiv 3$  that has no counterpart.

Direct computation refutes this conjecture.

**Observation 5.1** (Mod-5 census,  $N = 10^6$ ). The mod-5 distribution of the 37,915 witnesses in A002822 up to  $N = 10^6$  is:

|                |        |                  |
|----------------|--------|------------------|
| Class 0 mod 5: | 12,565 | (33.14%)         |
| Class 1 mod 5: | 1      | (witness 1 only) |
| Class 2 mod 5: | 12,690 | (33.47%)         |
| Class 3 mod 5: | 12,659 | (33.39%)         |
| Class 4 mod 5: | 0      | (0%)             |

Class 3 minus class 2 excess:  $-31$  (class 2 leads). Chi-squared test on classes  $\{0, 2, 3\}$ :  $\chi^2 = 0.67$ ,  $p = 0.72$ . Binomial test for class 3 > class 2:  $p = 0.58$ .

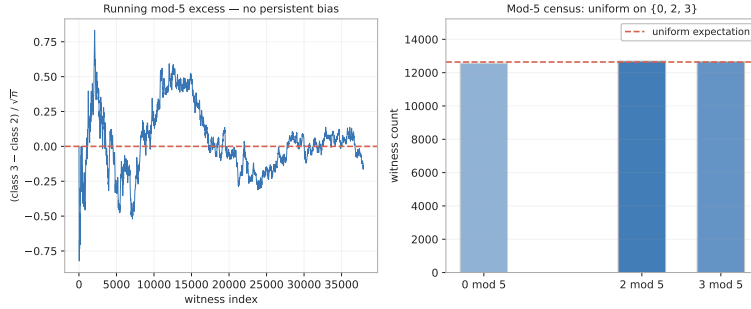


FIGURE 5. Left: mod-5 census of the 37,915 witnesses in A002822 up to  $N = 10^6$ . Classes 1 and 4 are structurally empty; classes 0, 2, 3 carry equal weight ( $\chi^2 = 0.67$ ,  $p = 0.72$ ). Right: running excess (class 3 minus class 2) oscillates around zero with no persistent drift. No Chebyshev-style bias is present.

The distribution over the three active classes is consistent with uniform ( $p = 0.72$ ). The running excess (cumulative count of class 3 minus class 2

as witnesses are scanned in order) oscillates around zero with standard deviation 25 and final value  $-31$ . There is no evidence of a persistent Chebyshev-style bias.

The stalling experiment (Observation 2.3) separately confirms that witness 1's structural role is real — but it manifests as load-bearing bridging, not as a bias in the residue distribution of the witnesses it generates.

**Remark 5.2.** The null result is informative. The partial semigroup theorem (Theorem 2.1) establishes that  $\mathcal{W} \subseteq \{0, 2, 3\} \pmod{5}$  — a genuine hard constraint on the *support* of the distribution. The null result establishes that the *distribution within the support* is uniform. The two facts are compatible and together give a complete picture: the mod-5 structure of  $\mathcal{W}$  is exactly what the partial semigroup theorem predicts, and no further arithmetic bias is visible at this scale.

## 6. SUMMARY OF THE EVIDENCE

The following table organises the heuristic and computational evidence by type and strength.

| Evidence                                                                          | Tier                          | Supports                             |
|-----------------------------------------------------------------------------------|-------------------------------|--------------------------------------|
| $\mathcal{W} \subseteq \{0, 2, 3\} \pmod{5}$ ; forbidden pairs (2, 2) and (3, 3)  | Proved (unconditional)        | Constrains which pairs can bridge    |
| $\text{WDC} \Rightarrow \text{GC}$                                                | Proved (unconditional)        | Structural coherence of WDC          |
| GC verified to $10^5$ witnesses                                                   | Computational                 | Indirect evidence for WDC            |
| Witness 1 stalling: seed $\{2, 3\}$ collapses from 37,915 to 7 witnesses          | Computational (unconditional) | Structural role of $\omega = 1$      |
| BM non-halting to $2 \times 10^{10}$ ; halting condition explicitly characterised | Computational                 | BC (and WDC) at large scale          |
| Survival rate tracks HL prediction across all 23 sieve generations                | Computational                 | Additive dynamics match analytic law |
| Stalling probability $\rightarrow 0$ as $e^{-2C_2k/(\log 2M_k)^2}$                | Analytic (under HLC)          | Confirms BC from below               |
| $D(w) \sim 4C_2^2w/(\log w)^4$ ; $\Pr[w \in \mathcal{X}] \approx e^{-D(w)}$       | Analytic (under HLC)          | Confirms BC from below               |
| Dubner: no exceptions beyond $w = 701$ to $3.3 \times 10^9$                       | Computational                 | BC empirical record                  |
| Mod-5 census ( $N = 10^6$ ): distribution uniform on $\{0, 2, 3\}$                | Computational (null result)   | No Chebyshev-style bias              |

The unconditional and computational tiers are the most significant. The partial semigroup theorem gives the framework; the BM non-halting evidence operates within that framework and makes the halting failure mode concrete. The HLC-based estimates in Section 4 confirm from below that BC is not asking for anything surprising — but BC does not need HLC to be true, and its credibility does not depend on HLC’s status.

The honest summary is: BC and WDC are overwhelmingly supported by the available evidence. No counterexample has been found; the structural prerequisites for failure are explicitly characterised and absent; the additive dynamics match analytic predictions at every scale examined. The gap between this evidence and a proof remains the central open problem. The

Structural Conjecture of [1] proposes that PNT is the missing link; proving it is the invitation extended to the analytic community.

#### ACKNOWLEDGEMENTS

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#### REFERENCES

- [1] J. Seymour, *A Structural Hypothesis for the Infinitude of Twin Primes*, Wild Duck Theories, May 2026. <https://wildducktheories.github.io/twin-primes/papers/structural/wdt-structural-conjecture.pdf>
- [2] J. Seymour, *How AI Was Used: A Transparency Statement for “A Structural Hypothesis for the Infinitude of Twin Primes”*, Wild Duck Theories, May 2026. <https://wildducktheories.github.io/twin-primes/papers/ai-usage/wdt-ai-usage.pdf>
- [3] F. Balestrieri, *Some remarks on the twin prime problem*, arXiv:1106.6050 (2011).
- [4] H. Dubner, *Twin prime statistics*, Journal of Integer Sequences **2** (1999), Article 99.1.1.
- [5] G. H. Hardy and J. E. Littlewood, *Some problems of ‘Partitio Numerorum’; III: On the expression of a number as a sum of primes*, Acta Mathematica **44** (1923), 1–70.
- [6] J. Friedlander and H. Iwaniec, *The polynomial  $X^2 + Y^4$  captures its primes*, Annals of Mathematics **148** (1998), 945–1040.

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